

Solutions to Review of JEE MAIN - 3 | JEE - 2022

PHYSICS

SECTION-1

$$1.(C) \quad \frac{mv^2}{R} = N \cos \theta$$

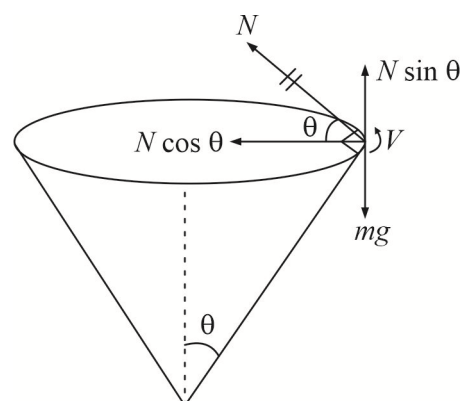
$$mg = N \sin \theta$$

$$\text{Divide } \cot \theta = \frac{v^2}{Rg}$$

$$\text{Also, } \cot \theta = \frac{h}{R}$$

$$\therefore v^2 = gh$$

$$v = \sqrt{gh} = \sqrt{9.8 \times 0.2} \text{ m/s} \Rightarrow v = 1.4 \text{ m/s}$$



$$2.(A) \quad F = -\frac{dU}{dr} = -6r^2 \Rightarrow \vec{F} = -6r^2 \hat{r}$$

$$F = \frac{mv^2}{r} \Rightarrow 6r^3 = mv^2 \Rightarrow K.E. = \frac{mv^2}{2} = 3r^3$$

$$P.E. = 2r^3$$

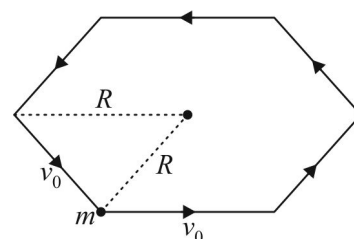
$$T.E. = 5r^3$$

$$\therefore T.E. = 5 \times 125 \text{ J} = 625 \text{ J}$$

$$3.(A) \quad \vec{\Delta P} = \vec{P}_f - \vec{P}_i$$

$$|\vec{P}_f - \vec{P}_i|$$

$$= \sqrt{(mv_0)^2 + (mv_0)^2 - 2(mv_0)^2 \cos 60^\circ} = mv_0 = 2mv_0 \sin \frac{\pi}{6}$$



$$4.(B) \quad ma = mg \sin \theta - \mu_k mg \cos \theta \Rightarrow ma = mg \sin \theta - 2mg \sin \theta \quad \therefore a = -g \sin \theta$$

$$\text{Speed} = v - g \sin \theta \cdot t = v - g \sin \theta \cdot \frac{v}{2g \sin \theta}$$

$$\text{Speed} = \frac{v}{2}$$

$$5.(D) \quad \text{Speed of bob just before collision}$$

$$u = \sqrt{2gh} = \sqrt{2g\ell(1 - \cos 60^\circ)} = \sqrt{g\ell} = \sqrt{980 \times 980} = 980 \text{ cm/s}$$

Momentum conservation in horizontal:

$$mu = 2mv_2 - mv_1$$

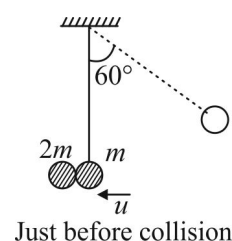
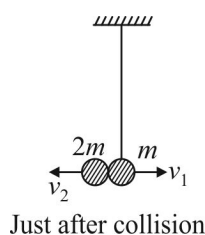
$$-v_1 + 2v_2 = u \quad \dots (1)$$

Also $e = 1$

$$v_1 + v_2 = u \quad \dots (2)$$

Solving (1) & (2)

$$v_2 = \frac{2u}{3}, \quad v_1 = \frac{u}{3} = \frac{980}{3} \text{ cm/s}$$



6.(B) $g - T = a$ (vertical block)
 $T - \mu g = a$ (horizontal block)
 Add, $(1 - \mu)g = 2a \Rightarrow 0.8g = 2a \Rightarrow a = 0.4g$

7.(A) $E_i + W_{\text{friction}} = E_f \Rightarrow \frac{Mv_0^2}{2} - \mu Mgx = +\frac{1}{2}kx^2$
 $\Rightarrow \frac{v_0^2}{2} - (0.9)(10)(0.4) = 50(0.4)^2 \Rightarrow v_0^2 = 23.2 \quad \therefore v_0 = \sqrt{23.2} \text{ m/s} = 4.8 \text{ m/s}$

8.(D) Loss in P.E. = Gain in K.E. $\Rightarrow mg(\ell - \ell \cos \theta_0) = \frac{mv^2}{2}$

$\therefore \frac{mv^2}{\ell} = 2mg(1 - \cos \theta_0) \dots\dots\dots \text{(i)}$

At lowest point, $T - mg = \frac{mv^2}{\ell} \dots\dots\dots \text{(ii)}$

By (i) & (ii),

$$T = mg(3 - 2 \cos \theta_0)$$

Also, $T < 4\mu mg \Rightarrow mg(3 - 2 \cos \theta_0) < 4\mu mg \quad \therefore \mu > \left(\frac{3 - 2 \cos \theta_0}{4} \right)$

9.(B) As $f_{\text{ext}}(\text{horizontal}) = 0$ & $u_{\text{cm}}(\text{horizontal}) = 0$

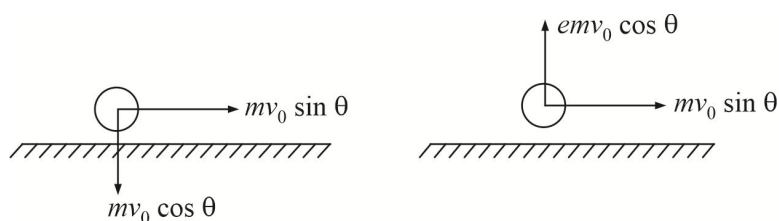
$$M_1 d_1 = M_2 d_2 \Rightarrow Mx = m(L - x) \Rightarrow (M + m)x = mL \quad \therefore x = \frac{mL}{M + m}$$

10.(C) acceleration $= a = -\frac{mg}{M + m} = -\frac{0.2}{0.7} \times 9.8 \text{ m/s}^2 = -2.8 \text{ m/s}^2 \quad \therefore v = v_0 + at = 7 - 2.8 \times 5 = -7 \text{ m/s}$
 Left is positive direction $\therefore v = 7 \text{ m/s}$, rightwards

11.(A) Limiting friction $= \mu mg = (0.4)(2)(9.8) \text{ N} = 7.84 \text{ N}$
 Block does not move
 Friction force = applied force $= F = 2.5 \text{ N}$

12.(D) Tangential acceleration $= a_t = \frac{v - u}{t} = \frac{26 - 17}{3} \text{ m/s}^2 \quad \therefore a_t = 3 \text{ m/s}^2$
 Tangential force $= F_t = ma_t = 30 \text{ N}$
 Centripetal force $= F_c = \frac{mv^2}{R} = 40 \text{ N}$
 Resultant force $= \sqrt{F_t^2 + F_c^2} = \sqrt{900 + 1600} \text{ N} = 50 \text{ N}$

13.(A) $\vec{P}_i = mv_0 \sin \theta \hat{i} - mv_0 \cos \theta \hat{j}$
 $\vec{P}_f = mv_0 \sin \theta \hat{i} + emv_0 \cos \theta \hat{j}$
 $\Delta \vec{P} = \vec{P}_f - \vec{P}_i = (1 + e)mv_0 \cos \theta \hat{j}$



Horizontal comp. remains same. Vertical comp. gets reversed & becomes e times.

$$14.(B) \quad \vec{F} = \frac{d\vec{P}}{dt} = 4 \cos t \hat{i} + 4 \sin t \hat{j}$$

$$\vec{F} \cdot \vec{P} = 16 \sin t \cdot \cos t - 16 \sin t \cdot \cos t = 0$$

$$\vec{F} \cdot \vec{P} = 0 \Rightarrow \vec{F} \perp \vec{P}$$

$$|\vec{P}| = 4 \text{ kg m/s} = \text{constant in magnitude}$$

$$15.(C) \quad \Delta P = \int F dt$$

For $0 - 2 \text{ sec}$:

$$P - 0 = \int_0^t 10t dt = 5t^2 \quad (\text{parabola upwards})$$

$$\text{At } t = 2, P = 5(4) = 20$$

For $2 - 4 \text{ sec}$:

$$P - 20 = \int_2^t (40 - 10t) dt$$

$$P = 20 + (40t - 5t^2) - (80 - 20)$$

$$P = -5t^2 + 40t - 40 \quad (\text{parabola downwards})$$

$$\text{At } t = 4, P = -80 + 160 - 40 = 40$$

$$16.(D) \quad W = \int \vec{F} \cdot d\vec{s} = \int (F_x dx + F_y dy) = \int (xy^2 dx + x^2 y dy) = \int xy(y dx + x dy)$$

$$= \int_0^B xy d(xy) = \left[\frac{x^2 y^2}{2} \right]_{(0,0)}^{(4m,0m)} = 0J; \quad W = 0J$$

$$17.(C) \quad \therefore a_{\text{pulley}} = \frac{a_1 + a_2}{2}$$

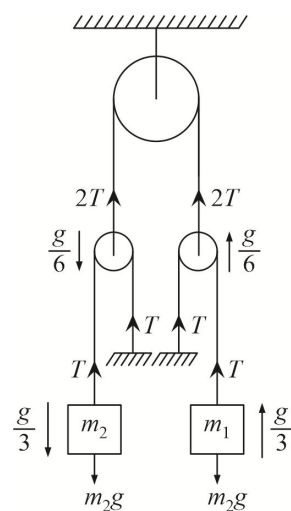
$$\therefore \text{Acceleration of } m_2 = \frac{g}{3} \downarrow$$

$$T - m_1 g = \frac{m_1 g}{3} \quad \text{for } m_1$$

$$\therefore T = \frac{4m_1 g}{3}$$

$$m_2 g - T = \frac{m_2 g}{3} \quad \text{for } m_2$$

$$\therefore T = \frac{2}{3} m_2 g \quad \therefore 4m_1 = 2m_2 \Rightarrow \frac{m_1}{m_2} = \frac{1}{2}$$



$$18.(D) \quad W_{\text{air}} + W_{\text{gravity}} = \Delta k \quad \Rightarrow \quad W_{\text{air}} + 0 = \frac{m}{2} \frac{v_0^2}{4} - \frac{m}{2} v_0^2 \quad \therefore \quad W_{\text{air}} = -\frac{3mv_0^2}{8}$$

$$19.(B) \quad F_x = -\frac{dU}{dx} = -10x + 20$$

$$F_x = 0 \quad \Rightarrow \quad x = 2m \text{ is equilibrium point}$$

$$\text{i.e.} \quad a = 0 \quad \Rightarrow \quad \text{Velocity is max. at } x = 2m$$

As $x = 2m$ is stable equilibrium point

$$20.(A) \quad a_{cm} = \frac{m_1 a_1 + m_2 a_2}{m_1 + m_2} = \frac{kx + F - kx}{m_1 + m_2} \quad \therefore \quad a_{cm} = \frac{F}{m_1 + m_2}$$

SECTION-2

$$1.(125) \quad F = V \frac{dm}{dt}$$

$$P = F \cdot V$$

$$P = v^2 \frac{dm}{dt}$$

$$\text{But,} \quad \rho A v = \frac{dm}{dt} \quad \Rightarrow \quad P = \frac{1}{\rho^2 A^2} \left(\frac{dm}{dt} \right)^3 \quad \therefore \quad \frac{dm}{dt} \text{ is 5 times} \quad \therefore \quad P \text{ is } (5)^3 \text{ i.e. 125 times}$$

$$2.(20) \quad \text{Loss in P.E.} = \text{Gain in K.E.} \quad \Rightarrow \quad mg \left(R - \frac{2R}{3} \right) = \frac{m}{2} V_B^2 \quad \Rightarrow \quad \frac{2gR}{3} = V_B^2$$

$$\therefore \quad V_B = \sqrt{\frac{2gR}{3}} = 20 \text{ m/s}$$

3.(60) Momentum conservation:

$$(2)(10) + (3)(0) = (2+3)V \quad \Rightarrow \quad 20 = 5V \quad \therefore \quad \text{Common velocity} = V = 4 \text{ m/s}$$

$$\text{Loss in kinetic energy} = KE_i - KE_f = \frac{1}{2}(2)(10)^2 - \frac{1}{2}(5)(4)^2 = 100J - 40J = 60J$$

$$4.(37.50) \quad \vec{P}_i = mv_0 \sin 45^\circ \hat{i} - mv_0 \cos 45^\circ \hat{j}$$

$$\vec{P}_f = mv_0 \sin 45^\circ \hat{i} + emv_0 \cos 45^\circ \hat{j}$$

$$v_i = v_0, v_f = \sqrt{\left(\frac{v_0}{\sqrt{2}} \right)^2 + \left(\frac{ev_0}{\sqrt{2}} \right)^2}$$

$$v_f = \frac{v_0}{\sqrt{2}} \sqrt{1 + \frac{1}{4}} = \frac{\sqrt{5}v_0}{2\sqrt{2}}$$

$$\text{Loss in K.E.} = \frac{m}{2} v_0^2 - \frac{m}{2} \cdot \frac{5v_0^2}{8} = \frac{3mv_0^2}{16}$$

$$\text{Percentage loss} = \left(\frac{3mv_0^2 / 16}{mv_0^2 / 2} \right) \times 100\% = \frac{300}{8}\% = 37.5\%$$

5.(3) Let the particle separate from the hemisphere at angular position θ .

$$mg \cos \theta - N = \frac{mv^2}{R}$$

Putting $N = 0$, $v^2 = gR \cos \theta$

Also, energy conservation gives $\frac{1}{2}mv^2 = mgR(1 - \cos \theta)$

$$\frac{gR \cos \theta}{2} = gR(1 - \cos \theta)$$

$$\frac{1}{2} \cos \theta + \cos \theta = 1 \quad \Rightarrow \quad \cos \theta = \frac{2}{3}$$

Let O be the origin

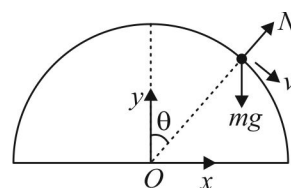
Coordinate of particle at this moment $= R \sin \theta, R \cos \theta = \left(\frac{\sqrt{5}}{3}R, \frac{2}{3}R \right)$

Coordinate of C.M of hemisphere $= \left(0, \frac{R}{2} \right)$

$$\therefore x_{\text{cm}} = \frac{m \left(\frac{\sqrt{5}}{3}R \right) + 4m(0)}{5m} = \frac{\sqrt{5}}{15}R$$

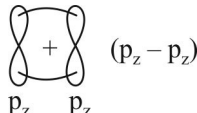
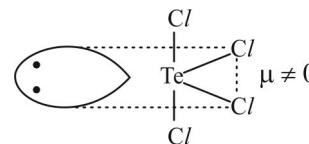
$$y_{\text{cm}} = \frac{m \left(\frac{2}{3}R \right) + 4m \left(\frac{R}{2} \right)}{5m} = \frac{8}{15}R$$

$$\therefore \text{Distance of C.M from } O = \sqrt{\left(\frac{\sqrt{5}}{15}R \right)^2 + \left(\frac{8}{15}R \right)^2} = \frac{\sqrt{69}R}{15} \quad \therefore \alpha = 3$$



CHEMISTRY

SECTION-1

1.(D) $\text{Ca}^{2+}\text{C}_2^{2-}$ 2.(A) π molecular orbital $p_z - p_z$ 3.(C) Hybridization of Cl in ClO_4^- $H = \frac{1}{2}[7+0-0+1] = \frac{8}{2} = sp^3$ Hybridization of O in OF_2 $H = \frac{1}{2}[6+2-0+0] = \frac{8}{2} = sp^3$ 4.(D) Size of $\text{Li}^+(\text{g})$ is smaller than $\text{H}^-(\text{g})$ & $\text{Na}^+(\text{g})$ due to effective nuclear charge. Size of $\text{Li}^+(\text{aq})$ is greater due to hydration. Same with F^- .5.(C) Bond order of $\text{O}_2 = 2$, bond order of $\text{O}_3 = 1.5$ and bond order of $\text{O}_2^{2-} = 1$ 6.(B) TeCl_4 have sea-saw shape i.e. distorted geometry $\therefore$ it will have permanent dipole moments.

7.(C) Formal charge = (valence electron) - (Non-bonding electron) - (1/2 bonding electron)

In H_2SO_4 FC on S = $6 - 0 - 8/2 = +2$ In H_3PO_4 FC on P = $5 - 0 - 8/2 = +1$

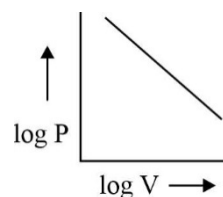
8.(B) According to ideal gas equation,

$$d = \frac{PM}{RT} \quad \therefore \quad d \propto \frac{P}{T}$$

Among the options, $\frac{P}{T}$ is highest for 2 atm and 273K.9.(B) According to kinetic theory, average kinetic energy $(E) = \frac{3}{2} k_B T$ Where, k_B is Boltzmann's constant. Since, it is independent of molar mass, it will be same for O_2 and H_2 at a given temperature.10.(B) According to Boyle's law, $PV = \text{constant}$

$$\therefore \log P + \log V = \text{constant}$$

$$\log P = -\log V + \text{constant}$$

Hence, the plot of $\log P$ Vs $\log V$ is straight line with negative slope.

11.(C) The expression of root mean square speed and average speed are

$$u_{\text{rms}} = \sqrt{\frac{3RT}{M}} \quad \text{and} \quad u_{\text{av}} = \sqrt{\frac{8RT}{\pi M}}$$

Hence,

$$\frac{u_{\text{rms}}}{u_{\text{av}}} = \left[\frac{3RT/M}{8RT/\pi M} \right]^{1/2} = \sqrt{\frac{3\pi}{8}}$$

12.(C) The mass of gas remains constant inside the bubble

$$\therefore \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\text{or, } \frac{V_2}{V_1} = \frac{P_1 T_2}{P_2 T_1}$$

$$\text{Given: } P_2 = 1 \text{ bar} \quad T_2 = 298 \text{ K}$$

$$P_1 = 1.5 \text{ bar} \quad T_1 = 288 \text{ K}$$

$$\frac{V_2}{V_1} = \frac{1.5 \text{ bar} \times 298 \text{ K}}{1 \text{ bar} \times 288 \text{ K}} = 1.6$$

$$\text{or, } V_2 = 1.6 V_1$$

$\therefore$ Volume increase by a factor of 1.6.

13.(A) From the graph we can see the correct order of pressures $p_1 > p_3 > p_2$

14.(B) Avogadro's hypothesis states that equal volumes of all gases under the same conditions of temperature and pressure contain equal number of moles or molecules.

1 gram atom = 1 mole.

According to ideal gas equation,

$$PV = nRT$$

$$\text{or } V = \frac{nRT}{P} = \frac{1 \text{ mole} \times 0.082 \text{ L atm K}^{-1} \text{ mole}^{-1} \times 273 \text{ K}}{1 \text{ atm}}$$

$$= 22.4 \text{ L}$$

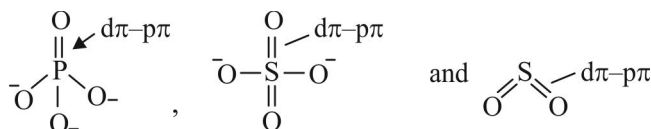
$\therefore$ 1 mole of any gas will occupy 22.4 L at STP.

15.(D) Let the mass of methane and oxygen = m gm.

Mole fraction of O_2

$$= \frac{\text{Moles of } O_2}{\text{Moles of } O_2 + \text{Moles of } CH_4} = \frac{m/32}{m/32 + m/16} = \frac{m/32}{3m/32} = \frac{1}{3}$$

16.(D) $d\pi - p\pi$ bond present in



$$\text{17.(A) Mass of Fe} = 100 \times \frac{0.33}{100} = 0.33 \text{ g}$$

$$\therefore \text{Moles of Fe} = \frac{0.33}{56} = 5.89 \times 10^{-3} \text{ mole}$$

18.(B) XeO_2F_2 have hybridisation sp^3d i.e. five orbital total

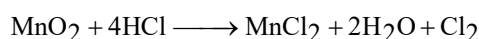
$\therefore$ have 20% d character. 20% s and 60% p-character

19.(B) Let V_{mL} of acid is required for neutralisation. Using equation of neutralisation we have:

$$N_a V_a = N_b V_b$$

$$\Rightarrow 0.5 \times V_{\infty} = 0.1 \times 500 \Rightarrow V_{\infty} = 100 \text{ cc} = 100 \text{ mL}$$

20.(D) Fact (in ice one molecule, of H_2O is surrounded by four molecules of H_2O with the help of Hydrogen bond.

SECTION-2**1.(50)**

From Stoichiometry 1mole 4moles 1 mole 2 moles 1 mole

Initially (Given) 2 moles 4 moles

[For 2 moles of MnO_2 , 8 moles of HCl are needed, but only 4 moles of HCl are given. Hence, HCl is the limiting reagent]

Finally 22.4L at STP (Theoretical yield) of Cl_2

$$\% \text{ yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100 \Rightarrow \% \text{ yield of } \text{Cl}_2 = \frac{11.2}{22.4} \times 100 = 50\%$$

2.(6) Molarity = 4.8 M

Let volume of solution = 1000mL = Volume of solvent

Mass of solvent = $1000 \times 0.8 = 800 \text{ g}$

Since, molarity of solution is 4.8 M

$\therefore n_{\text{solute}} = 4.8 \text{ mole}$

$$\text{Molality}(m) = \frac{4.8}{800/1000} = 6m.$$

3.(9) CN^- , H_2 , Li_2 , Be_2 , C_2 , N_2 , NO^+ , O_2^{2-} and F_2 are diamagnetic according to molecular orbital theory.**4.(5)** $\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

$$\text{No. of mole of } \text{CO}_2 = \frac{1 \times 0.448}{0.0821 \times 273} = 2 \times 10^{-2}$$

$$0.8n = 2 \times 10^{-2}$$

$$\Rightarrow n = \frac{2 \times 10^{-2}}{0.8} = 0.025$$

$$m_{\text{CaCO}_3} = 2.5 \text{ gm (pure)}$$

Mass of CaCO_3 sample = 5 gm.

5.(5) Amphoteric species which behave like both acid and base. They can donate or accept a proton.

MATHEMATICS

1.(A) 'CROMA'

ACMOR

Ans. (A)

(i) $A _ _ _ _ 4!$

(ii) $CA _ _ _ 3!$

$CM _ _ _ 3!$

$CO _ _ _ 3!$

(iii) $CRA _ _ 2!$

$CRM _ _ 2!$

(iv) $CROA _ 1!$

(v) $CROM.... 24 + 18 + 4 + 1 + 1 = 48$

2.(C) $\sum_{r=1}^{180} \cos r = \cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \cos 89^\circ + \cos 90^\circ$

$\cos 179^\circ + \cos 178^\circ + \cos 177^\circ + \cos 91^\circ + \cos 180^\circ$

$= \cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \cos 89^\circ + 0 - \cos 1^\circ - \cos 2^\circ - \cos 3^\circ \cos 89^\circ - 1 = -1$

3.(B) $x + y + z + \lambda = 3$

${}^{3+4-1}C_{4-1} = {}^6C_3 = \frac{6.5.4}{1.2.3} = 20$

4.(B) $P_n = \cos^n \theta + \sin^n \theta \Rightarrow P_n - P_{n-2} = \cos^n \theta + \sin^n \theta - \cos^{n-2} \theta - \sin^{n-2} \theta$
 $= \cos^{n-2} \theta (\cos^2 \theta - 1) + \sin^{n-2} \theta (\sin^2 \theta - 1) = \cos^{n-2} \theta (-\sin^2 \theta) + \sin^{n-2} \theta (-\cos^2 \theta)$
 $= -\sin^2 \theta \cos^2 \theta (\cos^{n-4} \theta + \sin^{n-4} \theta) = -\sin^2 \theta \cos^2 \theta P_{n-4}$

5.(C) No. of diagonals ${}^nC_2 - n = {}^8C_2 - 8 = 20 = m$

Now $20 = 2^2 \times 5^1$

No. of Divisors $3 \times 2 = 6$

6.(A) Gap method

$\times \times \times \times \times \times \times \times$

$7! \times {}^8P_3$

(Arrangement of Boys) (Arrangement of 3 girls in 8 Gaps)

7.(D) ${}^{n^2-n}C_2 = {}^{n^2-n}C_{10}$

$n^2 - n = 2 + 10$

$n^2 - n - 12 = 0$

$n^2 - 4n + 3n - 12 = 0$

$(n - 4)(n + 3) = 0$

$n = -3, 4$

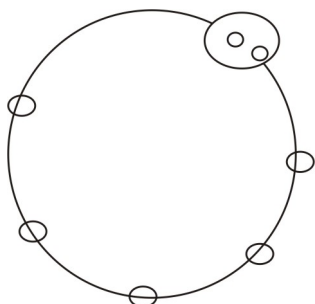
8.(C) No. of squares in a chess board $= 1^2 + 2^2 + 3^2 + + 8^2 = \frac{8 \times 9 \times 17}{6} = 204 = 2^2 \times 3 \times 17$

9.(B) Number of 4 digits odd numbers with the digits 0, 1, 2, 3, 4, 5 is:

$$4 \times 4 \times 3 \times 3 = 144$$

'0' cannot be on this place, last place can be filled with three ways.

10.(A)



$$(6-1)! \times 2! = 5! \times 2! = 120 \times 2 = 240$$

11.(C) JODHPUR

'PUR' always remains together

PUR J O D H

1 2 3 4 5

$$5! \times 3! = 120 \times 6 = 720$$

12.(C) P U R Never comes together

Total no. of arrangements – No. of arrangements in which 'PUR' together

$$= 7! - 720 = 720 \times 7 - 720 = 720 \times 6 = 4320$$

13.(B) $3^6 - {}^3C_1 \cdot 2^6 + {}^3C_2 \cdot 1^6 = 729 - 192 + 3 = 732 - 192 = 540$

14.(C) $\cos^2 \theta, \sec^2 \theta$ both are greater than or equals to 1, so roots should either 1 or greater.

(A) $\times$ $x^2 - 2x + 2 = 0$ roots imaginary as $D < 0$

(B) $\times$ $x^2 - 3x + 3 = 0$ roots imaginary as $D < 0$

$$(C) x^2 - 5x + 5 = 0 \quad \frac{5 \pm \sqrt{25 - 20}}{2}$$

$$= \frac{5 \pm \sqrt{5}}{2} \text{ both roots are more than}$$

(D) $x^2 + 4x - 4 = 0$ sum and products of roots negative.

15.(B) $a = \sin 170^\circ + \cos 170^\circ = \sin 10^\circ - \cos 10^\circ = \sin 10^\circ - \sin 80^\circ$

$$a < 0$$

16.(C) $A = \tan 6^\circ \tan 42^\circ$

$$B = \cot 66^\circ \cot 78^\circ$$

$$\text{Now } \tan(60^\circ - \theta) \tan \theta \tan(60^\circ + \theta) = \tan 3\theta$$

$$\theta = 18^\circ \quad \tan 42^\circ \tan 18^\circ \tan 78^\circ = \tan 54^\circ \dots\dots\dots(i)$$

$$\theta = 6^\circ \quad \tan(54^\circ) \tan 6^\circ \tan 66^\circ = \tan 18^\circ \dots\dots\dots(ii)$$

Multiply (i) and (ii)

$$\tan(42^\circ) \tan 18^\circ \tan 78^\circ \times \tan 54^\circ \tan 6^\circ \tan 66^\circ = \tan 54^\circ \times \tan 18^\circ$$

$$\tan 6^\circ \tan 42^\circ = \cot 66^\circ \cot 78^\circ$$

$$A = B$$

17.(A) $\cos 52^\circ + \cos 68^\circ + \cos 172^\circ = 2 \cos(60^\circ) \cos\left(\frac{68^\circ - 52^\circ}{2}\right) + \cos(180^\circ - 8^\circ)$

$$= 2 \times \frac{1}{2} \cos 8^\circ - \cos 8^\circ = \cos 8^\circ - \cos 8^\circ = 0$$

18.(B) $\sin(e^x) = 2^x + 2^{-x}$

$$-1 \leq \sin(e^x) \leq 1$$

$$\frac{2^x + 2^{-x}}{2} \geq \sqrt{2^0}$$

$$2^x + 2^{-x} \geq 2$$

So, no solution possible

19.(A) $\tan \frac{\pi}{3} + 2 \tan \frac{2\pi}{3} + 4 \tan \frac{4\pi}{3} + 8 \tan \frac{8\pi}{3} = \sqrt{3} - 2\sqrt{3} + 4\sqrt{3} - 8\sqrt{3} = -5\sqrt{3}$

20.(C) $\left[\frac{75}{5}\right] + \left[\frac{75}{5^2}\right] + \left[\frac{75}{5^3}\right] + \dots = 15 + 3 + 0 = 18$

$$\text{So } \cos 36^\circ - \sin 18^\circ = \frac{\sqrt{5}+1}{4} - \frac{\sqrt{5}-1}{4} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

SECTION-2

1.(8) $9600 = 96 \times 100 = 32 \times 3 \times 2^2 \times 5^2 = 2^{5+2} \times 3^1 \times 5^2$

No. of perfect square divisors: $4 \times 1 \times 2 = 8$

2.(64) ${}^{12}C_2 - {}^3C_2 + 1 = \frac{12 \times 11}{1 \times 2} - \frac{3 \times 2}{1 \times 2} + 1 = 66 - 3 + 1 = 63 + 1 = 64$

3.(72) A L G E B R A

Arrangement of Vowels $\times$ Arrangements of consonants

$$= \frac{3!}{2!} \times 4! = 3 \times 24 = 72$$

4.(9) No. of De-arrangements is $4! \left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right] = 12 - 4 + 1 = 9$

5.(2) Consider : $\sin x (\sin x + \cos x) = n$

$$\Rightarrow \sin^2 x + \sin x \cdot \cos x = n \Rightarrow \frac{1 - \cos 2x}{2} + \frac{\sin 2x}{2} = n \Rightarrow \sin 2x - \cos 2x = 2n - 1$$

$$\text{As } -\sqrt{2} \leq \sin 2x - \cos 2x \leq \sqrt{2} \Rightarrow -\sqrt{2} \leq 2n - 1 \leq \sqrt{2} \Rightarrow \frac{1 - \sqrt{2}}{2} \leq n \leq \frac{1 + \sqrt{2}}{2} \Rightarrow n = 0, 1.$$